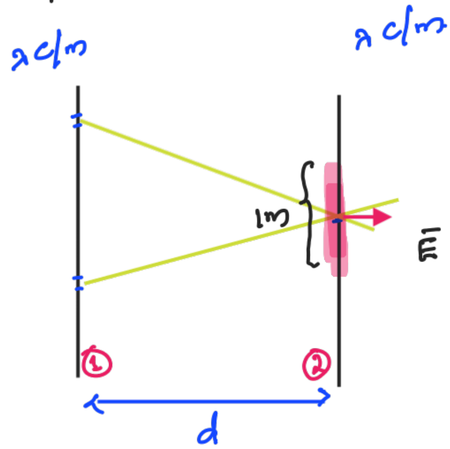
Example: Field due to two line charges:



Force per meter excerted in line charge② due to line charge①.

$$\vec{E} = \frac{\hat{r}\lambda}{2\pi\epsilon_0}\frac{1}{r}$$

$$= \frac{\hat{r}\lambda}{2\pi\epsilon_0}\frac{1}{d}.$$

This Electric field acts on 1m of charge in ②

$$\text{Force}/m = \lambda\vec{E}$$

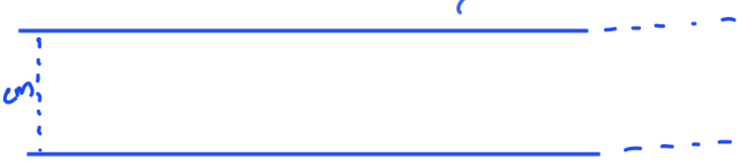
$$= \hat{r}\frac{\lambda^2}{2\pi\epsilon_0 d}$$

* It means the force due to these infinite line charge is infinity.
* In practical situations, infinite line charge is not possible. However approximate effect can be observed or realized.

Example: overhead power line: $d \approx$ cm;

AC freq ~50Hz; ⇒ $\lambda \sim \frac{c}{f} \approx 6000$ km.

100 km. $L \ll \lambda$ (quasi-static.)



≡ two infinite line charge.

Tx lines in ckts?

Coulomb's law:

$$\bar{F} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\bar{r}')\, dV' \,(\bar{r}-\bar{r}')}{|\bar{r}-\bar{r}'|^3}$$ ← Require to know exactly where the charge density everywhere & charge distribution.

* Mostly in electromagnetics, $\bar{r} \rightarrow$ observer point, $\bar{r}' \rightarrow$ source.

Usually we do not know where the charge is!!

Example: A metallic ball and if we put 1C charge on this ball. As it is a conductor, charges are free to move around and hence we do not know exact location of this 1C charge; only we know is the average charge and the shape of the conductor.

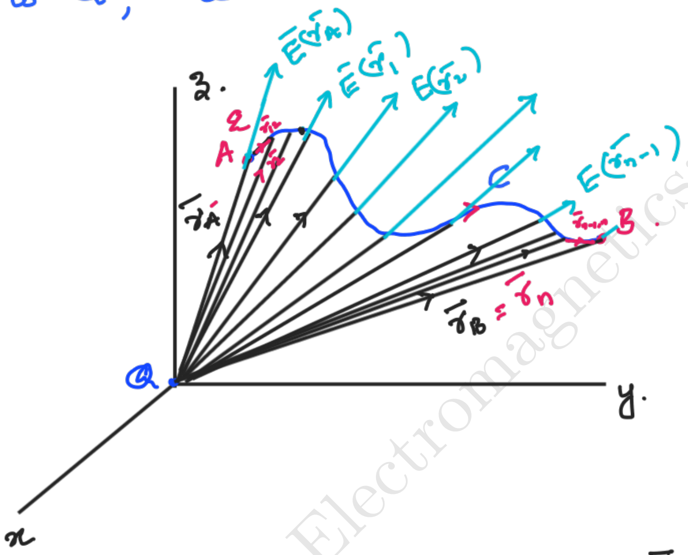
⇒ Coulomb's Eqn cannot be used

Very important component in Electrical engineering: Capacitor. 2 metallic plates separated by a distance. How will we find electric fields of such problems?

Work done inorder to move a charge

General Eqn : $dW = \bar{F} \cdot \bar{ds}$ } Usually it denotes kinetic energy.
Workdone = - Change in potential energy. (Law of energy conservation)

If 'q' is moved from A to B through 'C' in the presence of $\bar{E}$ due to 'Q', what is the workdone if it exist?



$\bar{r}_A = \bar{r}_1$
$\bar{r}_B = \bar{r}_n$

* Charge moves here not parallel to $\bar{E}$ (force $= q\bar{E}$)

Work done $= q\left[\bar{E}(\bar{r}_1) \cdot \bar{r}_{12} + \bar{E}(\bar{r}_2) \cdot \bar{r}_{23} + \dots E(\bar{r}_{n-1}) \cdot \bar{r}_{n-1,n}\right]$

When,
$d\bar{r} = \bar{r}_2 - \bar{r}_1 = \bar{r}_{12}$
$= \bar{r}_n - \bar{r}_{n-1} = \bar{r}_{n-1,n}$
& $\vec{dr} \to 0$

$= q\int_C \bar{E}(\bar{r}) \cdot \bar{dr}$ ← path integral.
(line integral.)

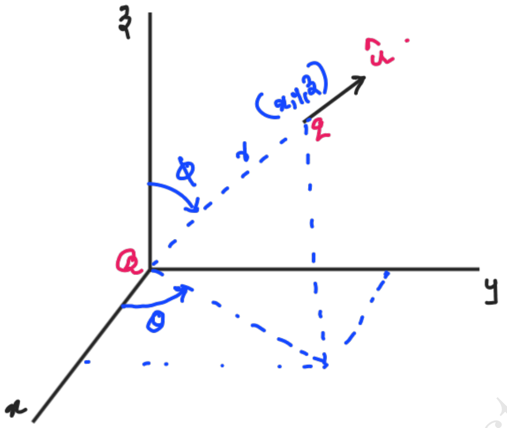
$= \int_C \bar{F}(\bar{r}) \cdot \bar{dr}$

↳ Try to remember tangential form of line integral (where unit vector was tangent vector) in Vector calculus - EE1203.

$$\boxed{W = \int_C \bar{F} \cdot \hat{t}\, ds}$$

Let us show the path independence in <u>Coulomb's force</u>.

Let a charge Q be fixed at the origin;
another charge q be situated at (x, y, z).



Coulomb force on 'q' is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{u} \quad ; \quad \bar{r} = xi + yj + zk.$$

$r = (x^2+y^2+z^2)^{1/2}$ (distance b/w q & Q)

$\hat{u}$ = unit vector pointing from Q to q

$$\hat{u} = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}}{r} = \frac{xi+yj+zk}{r}$$

$$\bar{F} = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^3} (xi + yj + zk)$$

$$\bar{F} = \langle F_x, F_y, F_z \rangle = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^3} \langle x, y, z \rangle$$

(Ref Lec 13 EE1203 - Vector Calculus)

x-component

$$\bar{F} \cdot \hat{t}\, ds = F_x dx + F_y dy + F_z dz$$

$$= \frac{qQ}{4\pi\epsilon_0} \frac{x\,dx + y\,dy + z\,dz}{r^3}$$

The trick is: use the relation

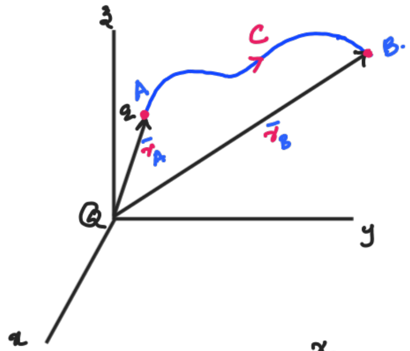
$$r^2 = x^2 + y^2 + z^2$$

Taking differentials in the equation

$$r\,dr = x\,dx + y\,dy + z\,dz$$

$$\Rightarrow \bar{F} \cdot \hat{t}\, ds = \frac{qQ}{4\pi\epsilon_0} \frac{r\,dr}{r^3} = \frac{qQ}{4\pi\epsilon_0} \frac{dr}{r^2}$$

If the charge 'q' moves from a point A at a distance r_1 from origin, to a point B at a distance r_2 over some path C connecting A & B; then.



$$\int_C \bar{F}\cdot d\bar{r} = \int_C \bar{F}\cdot\hat{t}\,ds = \frac{qQ}{4\pi\epsilon_0}\int_{r_A}^{r_B}\frac{dr}{r^2} = \frac{qQ}{4\pi\epsilon_0}\left(\frac{1}{r_A}-\frac{1}{r_B}\right)$$

$\Rightarrow$ not depending on the path 'C'.

ie; $\int_C \bar{F}\cdot\hat{t}\,ds$ with $\vec{F}$ (Coulomb force) is path independent

But $\bar{F} = q\vec{E}$

$$\underbrace{\int_C \bar{F}\cdot\hat{t}\,ds}_{\text{path independent}} = q\underbrace{\int_C \bar{E}\cdot\hat{t}\,ds}_{\text{path independent}}$$

So,
Total work done on charge 'q' is (= − Change in potential energy)

$$W = \int_C \bar{F}\cdot \bar{dr} = \frac{qQ}{4\pi\epsilon_0}\int_{r_A}^{r_B}\frac{dr}{r^2} = \frac{qQ}{4\pi\epsilon_0}\left[\frac{1}{r_A}-\frac{1}{r_B}\right]$$

In electricity and magnetism, we are more interested in potential energy gained by the charge.

$\Rightarrow$ PE gained by charge = − (work done on charge 'q')

$$= \frac{qQ}{4\pi\epsilon_0}\left(\frac{1}{r_B}-\frac{1}{r_A}\right)$$

initial point far away $\Rightarrow \frac{1}{r_A} \approx 0$.

$$= \frac{qQ}{4\pi\epsilon_0}\frac{1}{r_B}.$$

Potential energy gained by charge 'q' from far away to a point 'B' is,

$$P.E = \frac{qQ}{4\pi\epsilon_0}\frac{1}{r_B}.$$

We define electrostatic potential (ϕ) which is PE gained / Coulomb.

$$\boxed{\phi = \frac{P.E}{q} = \frac{Q}{4\pi\epsilon_0}\frac{1}{r}}.$$ ✓ --> Scalar & simpler function.

Compared to electric field

$$\boxed{\bar{E} = \frac{Q}{4\pi\epsilon_0}\frac{1}{r^3}\bar{r}.}$$ --- Vector function.

$$\phi = -\int_{\text{faraway}}^{r}\bar{E}\cdot\bar{dr} = \frac{Q}{4\pi\epsilon_0}\frac{1}{r}.$$

By superposition for cloud of charges in a region;

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\bar{r}')}{|\bar{r}-\bar{r}'|^3} (\bar{r}-\bar{r}')\, dv'.$$

Now potential due to each of $\rho(\bar{r})$ is,

$$\phi(\bar{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\bar{r}')\, dv'}{|\bar{r}-\bar{r}'|}$$

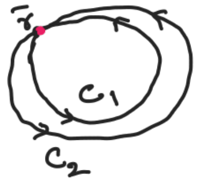
If we know ϕ on nearest points, we can find $\bar{E}$

$$\phi(\bar{r}+d\bar{r}) - \phi(\bar{r}) = -\left[\int_{\text{faraway}}^{\bar{r}+d\bar{r}} \bar{E}\cdot d\bar{r} - \int_{\text{faraway}}^{\bar{r}} \bar{E}\cdot d\bar{r}\right]$$

$$= -\int_{\bar{r}}^{\bar{r}+d\bar{r}} \bar{E}\cdot d\bar{r} \approx -\bar{E}\cdot d\bar{r}.$$

Important points on potential:

(1)
* $\phi(\bar{r}) = -\int_{\text{faraway}}^{\bar{r}} \bar{E}(\bar{r})\cdot d\bar{r}$ path independent line integral.

* 

Intuitively explain path independence requirement. If $V(\bar{r})$ was path dependent, different path gives different potential for the same point, which is not true.

(2) Advantage of potential formulation.

* Stokes' theorem;

$$\oint_C \bar{E}\cdot d\bar{r} = \iint_S (\nabla\times\bar{E})\cdot\hat{n}\,dS$$

$$\overset{\shortparallel}{0} \equiv \nabla\times\bar{E} = 0. \quad \text{(Conservative fields).}$$

* $\vec{E} \rightarrow$ electric field is a very special kind of a vector (as $\nabla\times\bar{E} = 0$)

(a) $\bar{E} = \hat{k}y \leftarrow$ cannot be a electric field vector

$$\nabla\times\bar{E} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & y \end{vmatrix}$$

$$= 1\,\hat{i}. \neq 0.$$

* This explains why a scalar field $\phi(\bar{r})$ is enough to completely determine vector field $\bar{E}$

$\nabla\times\bar{E} = 0$

$$\Rightarrow \frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}; \quad \frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z}; \quad \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}.$$

$\Rightarrow$ This helps us to reduce a vector problem to a scalar problem.

(3) The reference point for potential $\phi(\bar{r})$.

$$\phi'(\bar{r}) = -\int_{\substack{\text{faraway} \\ \text{new reference} \\ \text{point}}}^{r} \bar{E}\cdot d\bar{r} = \left[\underbrace{-\int_{\substack{\text{faraway new} \\ \text{point}}}^{\substack{\text{faraway old} \\ \text{point}}} \bar{E}\cdot d\bar{r}}_{K\text{ - Constant}} - \underbrace{\int_{\substack{\text{faraway} \\ \text{old point}}}^{r} \bar{E}\cdot d\bar{r}}_{\phi(\bar{r})} \right]$$

$$= K + \phi(\bar{r})$$

$$\phi'(\bar{r}_2) - \phi'(\bar{r}_1) = \phi(\bar{r}_2) - \phi(\bar{r}_1)$$ ← Change in reference does not affect the potential difference between two points.

* potential as such does not carry any real physical significance
  - → It is similar to altitude
    - ↳ Generally w.r.t sea level.
    - ↳ But we can make this reference other than sea level too, to any other reference.
    - ↳ Interestingly this will not change altitude difference between two places.
    - ↳ Similarly changing reference point does not vary potential difference between two points.

* For altitude we usually take sea level as reference; similarly for electric potential we take reference point as faraway ($\vec{r}_A \to \infty$); which is equivalent to selecting a zero potential point as reference.

  * One special case as in infinite plane charge case, where $\vec{E} = \frac{\sigma}{2\epsilon_0}\hat{n}$; even as $\vec{r}_A \to \infty$ potential blows up

$$\Phi(z) = -\int_{\infty}^{z} \frac{1}{2\epsilon_0}\sigma \, dz = \frac{-1}{2\epsilon_0}\sigma(z-\infty)$$

  * But in real life, always faraway point as reference ($\Phi(\text{faraway}) = 0$) will work out as there is no such infinite plane charge exist.

* Obeys superposition principle.

* Unit of potential : Nm/C or joules/Coulomb.

$$\underline{\underline{1\ J/C = 1\ \text{Volt}\ (V)}}$$

→ unit of $\vec{E}$ as $\underline{V/m}$.